CYBERSECURITY TRACK • PYTHON MODULE

Assignment 7

Automated Log Analyzer Tool (Extended Edition)

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1. Project Overview

The objective of this assignment is to develop a specialized Python script tailored for SOC analysts, which automatically reads web server log files line by line, applies Regular Expressions (Regex) to detect various attack attempts such as SQL Injection or Path Traversal, and prints out security alerts[cite: 1]. This extended version includes advanced argument parsing, severity classification, and multi-format reporting.

2. Development Steps

1 **Log File Reading**

Read text log files line by line using `with open` blocks[cite: 1].

2 **Regex Pattern Definition**

Define matching patterns utilizing the `re` module to detect suspicious text such as `UNION SELECT` or `../etc/passwd` [cite: 1].

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Log Iteration

Iterate through the logs and apply inspection patterns on each individual line[cite: 1].

Custom Alerting & Export

Print a customized security alert containing the detected attack type, line number, and attacker IP[cite: 1], and export results to JSON.

3. Technical Features & Utilities

Case Insensitivity: Use `re.compile(pattern, re.IGNORECASE)` to ignore uppercase and lowercase characters during inspection[cite: 1].

Direct Search Method: Use the `re.search()` method inside conditions for direct searching within lines[cite: 1].

4. Implementation Source Code

`log_analyzer.py`

Python 3.x

```
import re
import sys
import os
import time
import argparse
import json

class LogAnalyzer:
    def __init__(self, log_file_path, output_format=None):
```

```
        self.log_file_path = log_file_path
        self.output_format = output_format
        self.patterns = self._load_attack_patterns()

    def _load_attack_patterns(self):
        # Comprehensive signatures compiled with re.IGNORECASE
        return {
            "SQL Injection": {
                "regex": re.compile(r"(union\s+select|or\s+1=1|select\s+.*\s+from|drop\s+table|exec\s*\(|
                "severity": "HIGH"
            },
            "Path Traversal": {
                "regex": re.compile(r"(\.\./\.\./|\.\./etc/passwd|windows/win.ini|/etc/shadow)", re.IGNOR
                "severity": "CRITICAL"
            },
            "Cross-Site Scripting (XSS)": {
                "regex": re.compile(r"(
```